Understanding *Social Media Crisis*: A Data-driven Analysis of Online Problematic Behavior

JEANNE CHOI* and DONGJAE KANG*, KAIST, Republic of Korea
JOSEPH SEERING, KAIST, Republic of Korea

As social media platforms increasingly become primary channels for news consumption and dissemination, understanding problematic behaviors within these platforms becomes crucial for maintaining information integrity. While current HCI research provides insights into online problematic behavior, there is a gap in comprehending how these behaviors evolve within the dynamic context and nature of social media. To address this gap, our study introduces the concept of *Social Media Crisis*: events characterized by online problematic behavior that originates and evolves primarily within social media, requiring societal intervention. Through analysis of news articles from 2004 to 2023 utilizing GPT-4, we identify and examine 19 cases of *Social Media Crisis* to understand how platform dynamics and affordances influence information flow and crisis development. Our findings reveal patterns in how platform-specific features contribute to crisis evolution, particularly in the context of news dissemination and consumption. This research seeks to inform the design of safer social media platforms, enabling proactive measures to mitigate potential crises.

CCS Concepts: • **Human-centered Computing**; • **Human computer interaction(HCI); Empirical Studies in HCI**;

Additional Key Words and Phrases: Social Media, Online Problematic Behavior, Information Integrity, Platform Affordances, Crisis Management, Platform Design, Data-driven Analysis, Human-Computer Interaction

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1 Introduction

As reliance on social media grows, so too has the prevalence of online problematic behaviors, including misinformation [14, 20, 23], harassment [6], and other harmful actions that significantly impact news dissemination and consumption. These behaviors, when occurring within news-sharing contexts, can escalate to serious social problems [16], adding complexity and potentially leading to more severe consequences for information integrity. To address these risks in today's evolving news ecosystem, it is critical to understand how such behaviors evolve in specific social media contexts.

While HCI research has made progress in studying user behavior in social media [12, 17, 21, 22], one major limitation is the absence of comprehensive datasets that can help predict and prevent problematic behavior based on the surrounding social environment, particularly in social media contexts. To address this gap, we have identified a dataset of crises

*Both authors contributed equally to this research.

Authors' Contact Information: Jeanne Choi, jeannechoi@kaist.ac.kr; Dongjae Kang, jackkang3780@kaist.ac.kr, KAIST, Daejeon, Republic of Korea; Joseph Seering, seering@kaist.ac.kr, KAIST, Daejeon, Republic of Korea.

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endemic to social media within the modern news ecosystem, offering a focused lens to study how these crises originate, evolve, and ultimately require intervention.

To understand this situation with a comprehensive framework that considers interactions over time, we introduce the concept of a *Social Media Crisis*, which we define as: "*as an event or phenomenon characterized by online problematic behavior that originates and evolves primarily within social media, and has a societal impact that subsequently requires a timely action, mitigation, or societal intervention*". Here, we define online problematic behavior as a behavior that breaks the social harmony in the online space, such as harassment, misinformation, and aggressive reactions among a large number of users. *Social Media Crisis* typically involves polarized opinions, real-world implications, and challenges in content moderation by platforms. Also, *Social Media Crisis* often causes a rapid spread of information, which leads to widespread engagement of social media platform users and resulting in real-world consequences. This new concept emphasizes how severe events in social media uniquely emerge and escalate because of the dynamics and affordances specific to social media platforms, particularly in the context of news sharing and consumption.

Social Media Crisis shares similarities with traditional crises as they both have high levels of immediacy and severity [8]. However, it is distinct in how it progresses and is amplified within digital environments. Such crises incorporate the experiences of social media use during critical situations [1–3, 7, 10] and are intertwined with trust and safety issues [16, 18], both of which are central concerns in the evolving news ecosystem. By framing these incidents as *Social Media Crisis*, we gain a deeper understanding of their development, from their roots in problematic behavior to the following societal responses.

In this research, we identify how specific affordances of social media platforms contribute to these interactions, particularly in the context of news dissemination. By applying these findings, social media platforms can anticipate potential crises caused by specific affordances and take them into consideration during the design process, thereby contributing to the creation of a safer and more trustworthy social media environment.

2 Dataset and Method

2.1 Data Collection and Processing

We have collected data from news articles spanning 2004 to 2023 through TDM studio[1], a data mining solution platform that enables comprehensive news article analysis. The collection period begins from 2004, marking the emergence of Web 2.0 and the establishment of Facebook, which fundamentally transformed how news is shared and consumed online [19].

Our data sources (Table 1) include US Major Dailies [15], selected for their journalistic credibility, alongside representative news sources from Asia and European newsstream, regions with high social media penetration and diverse news consumption patterns. We also included technology magazines (Bloomberg Businessweek, Infoworld, Network World, Computerworld)[2] to capture how the technology sector perceives and reports on social media-related incidents. The data collection process utilized keywords including 'Social Media', 'Online', and names of prominent social media platforms for each year, identified through Google Trends[3] 'Online Community' category. This approach ensures we capture the evolution of platforms' roles in news dissemination over time.

[1]TDM studio (https://tdmstudio.proquest.com/home)
[2]Bloomberg Businessweek(https://www.bloomberg.com/businessweek), Infoworld(https://www.infoworld.com/), Network World(https://www.networkworld.com/), Computerworld(https://www.computerworld.com/)
[3]Google Trends (https://trends.google.com/trends/)

Table 1. News Publisher List by Publication Country and Publisher Title

Category	Publication Country	Publisher Title
U.S. Major Dailies	U.S.	Chicago Tribune Los Angeles Times New York Times Wall Street Journal The Washington Post
Asia News Publishers	China	Xinhua News Agency - CEIS People's Daily
	Singapore	The Straits Times
	India	The Hindustan Times The Times of India
	South Korea	The Korea Times Yonhap News Agency
	Japan	The Japan News
	Thailand	Asia News Monitor The Nation
	Asia-Wide	BBC Monitoring Asia Pacific BBC Monitoring Central Asia BBC Monitoring South Asia BBC Monitoring Newsfile BBC Monitoring Media
European News Publishers	UK	The Guardian The Daily Telegraph
	Ireland	Irish Times Irish Examiner
	France	Le Monde
	Germany	Die Tageszeitung Die Welt DPA International (English)
	Spain	El Pais
	Europe-Wide	BBC Monitoring European BBC Monitoring Former Soviet Union BBC Monitoring Newsfile BBC Monitoring Media

2.2 Identification of *Social Media Crisis*

Our analysis employed a three-stage filtering process to identify and verify social media crises:

(1) *Initial Event Identification*: We have applied topic modeling with BERTopic [5] to the collected dataset, then used GPT-4 [13] to filter the articles on those that are related to social media. We have validated the results against original news articles, and identified 303 initial events.
(2) *Candidate Selection*: Two of the first authors independently evaluated events for the presence of online problematic behavior, and selected the events that showed agreement between the authors. As a result, we have refined to about 60 candidate cases.
(3) *Final Social Media Crisis Verification*: In this step the events were evaluated based on three criteria: 1) Social media causality - the event's dependence on social media existence, 2) Stakeholder reaction - whether there is a

Table 2. *Social Media Crisis*

Number	Year	Case
1	2006	The Megan Meier Cyberbullying
2	2009	The Jang Ja-yeon Abuse and Exploitation Scandal with Online Defamation
3	2011	The Arab Spring Social Media Disinformation and Harassment
4	2011	The Sean Duffy Trolling and Cyber Harassment
5	2014	Harassment on minority groups in the online gaming society happens mainly with the hashtag of #Gamergate
6	2014	Facebook manipulated user data for social media research, without proper consent
7	2017	Kendall and Kylie Jenner get criticized on social media for their created T-shirts
8	2017	US President Trump uploaded Tweeting containing harassment towards other people
9	2018	Chrissy Teigen and Pizzagate: A Social Media Misinformation and Harassment
10	2018	Logan Paul's Video Backlash and Social Media Ethics Controversy/Discussion
11	2018	Cambridge Analytica Scandal
12	2018	Rumors based on fake videos and messages spread in WhatsApp kill 14 lives
13	2018	Harsh reports of men in an Instagram Account, Diet Madison Avenue
14	2018	Harassment towards Sushma Swaraj
15	2021	India's COVID-19 Vaccine Misinformation and Hesitancy
16	2022	Misinformation(or MDM) in the Russo-Ukrainian War
17	2022	Blocking and Platform Actions Regarding Russia and Ukraine
18	2022	The Lazada TikTok Ad Controversy and Royalist Backlash
19	2023	2023 Brazilian Congress attack, Insurrection

response from multiple stakeholders beyond immediate participants, and 3) Complete scenario – whether the event has a clear start-to-end progression. Through this step, we could result in 19 verified *Social Media Crisis*.

2.3 Analysis Method

We employed open coding [4] to analyze the progression and outcomes of each *Social Media Crisis*, particularly focusing on their impact on news dissemination and consumption. The first authors independently coded on the progression and outcomes of 5 cases, discussed the major themes discovered, and coded on the remaining cases based on the agreed code theme. Our coding framework organized around key features of online problematic behavior in news contexts, stakeholder roles and interactions, media platforms and their specific affordances, and platform characteristics contributing to crisis development.

Also, we referenced crisis management frameworks [9, 11] to code: (1) *Pre-crisis*: Causes and initial conditions, (2) *Crisis development*: Information flow and stakeholder responses, and (3) *Post-crisis*: Intervention strategies and societal impact. This analytical approach allowed us to understand how social media crises affect and are affected by changes in the the social media environment, enabling insights into platform design and crisis prevention.

3 Analysis

Platform Dynamics and Information Flow. Our analysis identified 19 significant social media crises that demonstrate how platform characteristics influence news dissemination and consumption (Table 2). These cases span from 2006 to 2023, showing the evolution of crisis patterns alongside platform development.

The modality analysis reveals that while text remains the primary medium across all cases, image and video content play increasingly crucial roles in crisis development, particularly in cases involving misinformation and harassment. This multi-modal nature of modern social media platforms significantly impacts how news and information spread during situations of *Social Media Crisis*.

Platform distribution shows Twitter (11 cases), Facebook (6 cases), and YouTube (4 cases) as primary crisis venues, with newer platforms like Instagram (3 cases) and Telegram (3 cases) emerging in recent cases. This evolution reflects changing patterns in news consumption and sharing behaviors.

Crisis Origins and Information Integrity. Our analysis reveals two primary categories of crisis origins. The first category is *Platform-Endemic Causes*, which includes direct online problematic behaviors such as harassment and defamation, misinformation spread within platform ecosystems, and controversial content triggering platform-wide discussions. The second category is *External Triggers with Platform Amplification*, which encompasses political events leading to online information manipulation, unethical platform practices affecting user trust, and crisis amplification through platform-specific features.

Stakeholder Dynamics in News Ecosystem. *Social Media Crisis* reveal complex interactions between multiple stakeholders: information producers including traditional media and individual creators, platform operators and moderators, information consumers and redistributors, and external stakeholders such as legal bodies and advocacy groups.

Platform Affordances and Crisis Evolution. Key platform characteristics influencing crisis development include *Content Creation and Distribution*, which involves anonymous posting reducing accountability, limited content verification processes, easy content creation and viral spread, and flexible content modification options.

Information Consumption Patterns are another characteristic, encompassing algorithm-driven content feeds, lack of transparency in content recommendation, and limited accountability in information sources.

The third characteristic is *Platform Structure Impact*, which includes decentralized information flow, networked user connections, and challenges in real-time content moderation.

Crisis Intervention and Aftermath. Intervention strategies varied in their normative strength. *High-normative Interventions* included legal actions and policy implementation, platform-level content moderation, and new platform guidelines. *Community-based Interventions* encompassed public accountability measures, user group agreements, and offline awareness campaigns.

The aftermath of these crises showed both constructive and harmful outcomes for the news ecosystem. *Constructive Outcomes* comprised enhanced platform accountability, improved information verification systems, and increased user awareness of misinformation. *Harmful Outcomes* consisted of escalated user conflicts, information polarization, and decreased trust in news sharing platforms which are mainly social media platforms.

4 Discussion

Our research introduces and analyzes the concept of *Social Media Crisis* within cases shown in the evolving news ecosystem, providing several key insights for understanding and addressing the challenges of modern news dissemination and consumption.

4.1 Theoretical and Methodological Contributions

Our work makes several contributions to understanding the changing news landscape. In this research, we develop a framework that introduces *Social Media Crisis* as a novel framework for understanding how problematic behaviors in social media affect news integrity. This is done by providing a structured approach to analyzing the evolution of *Social Media Crisis* in digital news spaces, and thus offering a comprehensive dataset of *Social Media Crisis* affecting news ecosystems.

Also, we provide design implications for social media platforms by identifying specific platform features that contribute to information crisis development. This highlights the role of platform affordances in shaping news consumption patterns, and especially suggests design considerations for creating a more resilient news sharing environments. Lastly, we demonstrate the value of using news source analysis through LLM for understanding social media dynamics by providing a systematic approach to identifying and analyzing platform-specific crises.

4.2 Possible Future Research Directions

While our research provides valuable insights, several areas merit further investigation. Beyond problematic behavior, our research explores how social movements like #BlackLivesMatter and #MeToo demonstrate social media's ability to facilitate important public discourse, which would be an important topic to be investigated solely. We also seek the need for future research on how platform research organizations' data manipulation raises questions about institutional power in digital news spaces, which were topics mentioned in the cases of data manipulation in *Social Media Crisis*. Lastly, some of the *Social Media Crisis* cases where external events trigger online responses require further examination, as it was not the cases that were focused in this research.

4.3 Design Implications for News Futures

Our findings suggest several considerations for future platform design. Crisis prevention features would need to be included through early warning systems for potential crises based on a thourough understanding on cases of *Social Media Crisis*, which leads to a need to an improved stakeholder communication channel specifically constructed for crisis prevention. Also, user education on potential harm regarding crisis they could face and awareness tools for users would help mitigating *Social Media Crisis*. Lastly, community engagement could be an important part for addressing problematic behavior, which could involve supporting positive social movements or facilitating constructive dialogue.

5 Conclusion

This research introduces the concept of *Social Media Crisis* as a framework for understanding how problematic online behaviors impact news integrity and consumption. Through analysis of 19 significant cases spanning 2006-2023, we identified key platform characteristics that influence crisis development, stakeholder dynamics within the news ecosystem, and intervention strategies of varying normative strength. Our findings reveal both constructive outcomes such as enhanced platform accountability or improved verification systems, and harmful consequences including user conflicts and information polarization resulting from these crises. These insights contribute to the theoretical understanding of digital news spaces while offering practical design implications for creating more resilient news-sharing environments through trust-building mechanisms, crisis prevention features, and community engagement tools. As social media continues to evolve, further research into positive platform uses, institutional power dynamics, and

granular interaction patterns will be essential for developing platforms that support responsible news dissemination while mitigating the risks of problematic behavior.

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